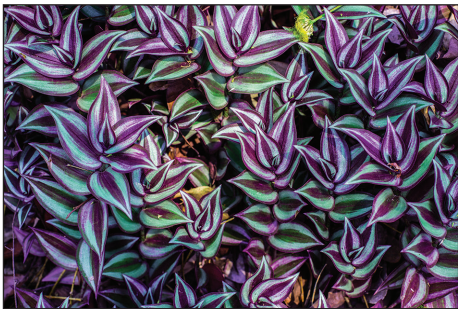


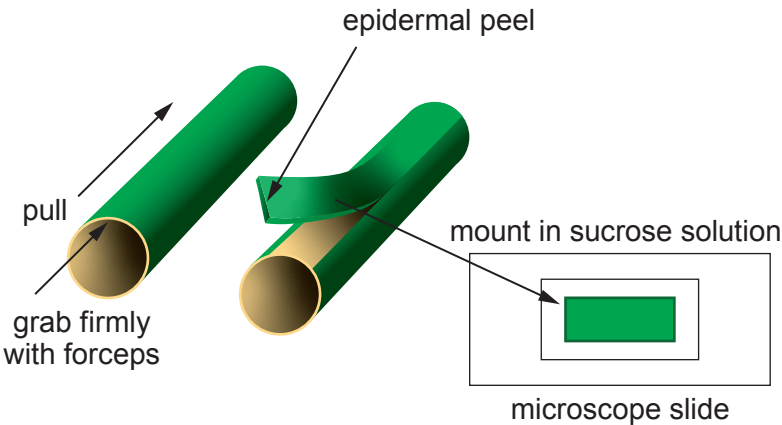
6. *Tradescantia*, shown in **Image 6.1**, is a house plant which is popular due to its ability to survive long periods without watering because its stems can store water.

Image 6.1



The solute potential of the cells of the stem was measured by using epidermal peels as shown in **Image 6.2**. Each peel was mounted on a different microscope slide in a different concentration of sucrose solution.

Image 6.2



After 10 minutes the peels were observed under a microscope. The number of plasmolysed and unplasmolysed cells in each sample was recorded. The results are shown in **Table 6.3**.

Table 6.3

Sucrose concentration/ mol dm ⁻³	Number of plasmolysed cells	Number of unplasmolysed cells	Total cells counted	Percentage of cells plasmolysed
0.0	0	165	165	0.0
0.2	20	174	194	10.3
0.4	90	105	195	46.2
0.6	87	69	156	55.8
0.8	76	54	130	58.4
1.0	172	73		

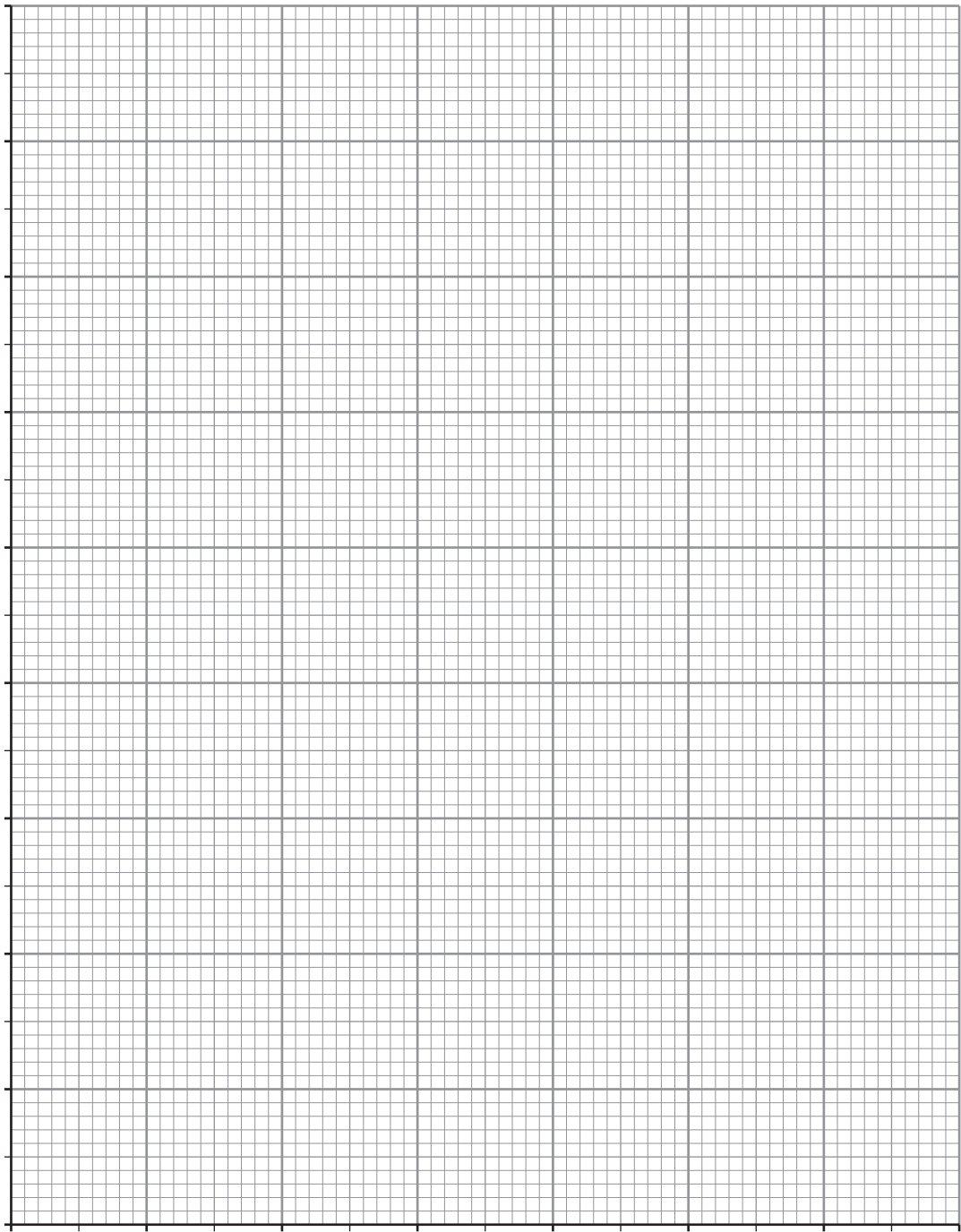
(a) (i) **Complete Table 6.3** for 1.0 mol dm⁻³ sucrose. [2]



(ii) Use the data in **Table 6.3** to draw a graph which shows the percentage of plasmolysed cells against sucrose concentration.

[4]

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- (iii) I. Use your graph and the formula below to estimate the solute potential of the solution when 50% of the *Tradescantia* cells are plasmolysed. [2]

Solute potential (Ψ_s) = - (sucrose concentration $\times$ 24.35)

Solute potential = kPa

- II. State the pressure potential of the cells in the tissue of *Tradescantia* at this solute concentration and identify the term that describes their condition. [2]

Pressure potential (Ψ_p) = kPa

Term =

- (b) Suggest how the experiment could be altered to improve:

- confidence in the results
- the accuracy of the estimated solute potential in *Tradescantia* cells. [2]

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